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Student Grade Predictor: Early Support Risk Classification

Introduction

Predicting possible academic risk before final results may help educators direct attention toward students who could benefit from additional support. This project explores whether selected study-habit, attendance, and school-support information can identify students who did not reach the passing threshold in mathematics. The analysis uses 395 records from the UCI Student Performance dataset, collected from two Portuguese secondary schools.

The purpose is to demonstrate a reproducible beginner machine-learning workflow. The model is an educational exercise and is not intended to assess or label individual students.

Problem Statement

The project classifies final mathematics grade G3 as at-risk/fail when it is below 10 on the dataset's 0-to-20 scale, and pass otherwise. The model uses study time, prior failures, school and family support, paid classes, activities, plans for higher education, internet access, absences, travel time, free time, going out, weekday and weekend alcohol-use indicators, and health rating.

G1 and G2 are excluded from the inputs because earlier-period grades are strongly related to G3 and could make prediction less useful as an early-support exercise. G3 is used only to define the target class. The dataset contains 130 at-risk/fail records and 265 pass records.

Results and Discussion

A stratified 80/20 split with random seed 42 produced 316 training records and 79 test records. Categorical inputs were one-hot encoded, and feature scaling was fitted on the training split only. A class-balanced, L2-regularized logistic regression model trained with gradient descent was compared with a majority-class baseline. The holdout results and confusion matrix are shown on the next page.

Confusion Matrix

The positive class is at-risk/fail. The matrix shows the logistic regression results on the 79-record test split.

Actual / Predicted	Pass / not at risk	At risk / fail
Pass / not at risk	39	14
At risk / fail	14	12

True negatives = 39; false positives = 14; false negatives = 14; true positives = 12.

Model Evaluation Code

```
weights, bias = fit_logistic_regression(X_train, y_train)
probability = sigmoid(X_test @ weights + bias)
prediction = (probability >= 0.5).astype(int)

Majority baseline: accuracy 0.671; at-risk recall 0.000; balanced accuracy 0.500
Logistic regression: accuracy 0.646; at-risk recall 0.462; balanced accuracy 0.599
```

The baseline has higher accuracy because pass is the more common class, but it misses all at-risk test records. Logistic regression finds 12 of 26 at-risk records and raises 14 false alarms. Its 0.462 recall is limited, so accuracy alone would give an incomplete view of the result. This is one small holdout split; scores may change with a different split.

Conclusion

This project demonstrates data preparation, binary classification, and evaluation against a simple baseline. The model improves balanced accuracy over the baseline (0.599 versus 0.500), but it misses more than half of the at-risk records in the test split. The result is modest and does not establish that any input causes a grade outcome. The model should not be used to label, rank, penalize, or make decisions about individual students.

Reference

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T. Dataset license: Creative Commons Attribution 4.0 (CC BY 4.0).